Sprint 1 – Spotify track dataset

In this sprint you will work with real world data crawled from Spotify. You will perform a thorough EDA and will then use the results to train a genre classifier.

Dataset

The dataset contains over 2 million records, each containing information about a single song. For each song, we have attributes such as the title, artist, release date, a genre label and several audio features such as “danceability” and “tempo”.

You can download the data here:
https://cloud.ilabt.imec.be/index.php/s/Lmp6WZJtpjXiqMY

Assignment

Your task is to develop a machine learning model to predict the genre for a new input song. You at least need to perform the following tasks:

- Perform a thorough EDA to get insight into the dataset and to identify potential issues. Clean up the data if needed.
- Follow all best practices in model validation: split your data in train, test, validation splits and use cross validation.
- Train at least three classifiers from different model families.
- Perform hyperparameter tuning to search for the optimal hyperparameters.
- Use good metrics to compare the performance of the models.
- Generate visualizations such as confusion matrices and discuss them.
- Identify what features are the most useful and re-train your best model only on those features.
- Analyze the results both quantitatively and qualitatively.

You are free to use any machine learning model and feature combination. You can use the dataset any way you want and are free to limit the scope of the assignment by for example using only a subset of the data, combining different classes, introducing new features, … but you cannot use any external data.

Submission and grading

Your solution will be graded in two steps:

- First, you submit a Jupyter notebook which contains all code to Ufora. Make sure to deliver a clean notebook that is organized in subsections that reflect the different steps of the workflow, and that contains sufficient explanations and discussions, not just documentation of the code. If the notebook is too large, you can submit different notebooks. **Also submit a pdf version of the notebook with only the relevant output included (i.e. no long outputs of training loops). This is the version you will use during the test.**

- During the test, you will be questioned on your approach and your findings, in a restricted setting (i.e. no internet access). **You need to bring a printed copy of the pdf you submitted.** This will be the only resource you can use. You can expect questions such as:
    - What problems did you find with the dataset.
    - How did you split the data and why?
    - What features did you use to train the model ? How did you preprocess them ?
    - What models did you use and why ?
    - What performance did you obtain ? Why this metric ?
    - Is this a good performance ?

  The goal of the test is not to question theory knowledge such as the inner workings of the models but to show that you reflected critically about the results and the different steps you took. Make sure you can motivate every decision.

Contact hours

25-sep	Course Intro / ML Pipeline	No session
2-okt	Training Models	Training Models + intro sprint 1
9-okt	Performance metrics	Workshop 1
16-okt	Decision Trees	PCA (Pieter)
23-okt	Clustering	Contact hours:
30-okt	Intro NN	**Test 1 (Written project defense)**
6-nov	CNN	Workshop 2a
13-nov	RNN	Workshop 2b
20-nov	Transformers	Contact hours
27-nov	GMM; MLE	Contact hours
4-dec	VAE, Diffusion, GAN	Contact hours
11-dec	**No class**	**Test 2 (Written project defense)**

Workshop, tests and contact hours take place in : GSCHB.2.035 and adjacent class rooms.